



Surface hydrolysis-anchored eugenol self-polishing marine antifouling coating

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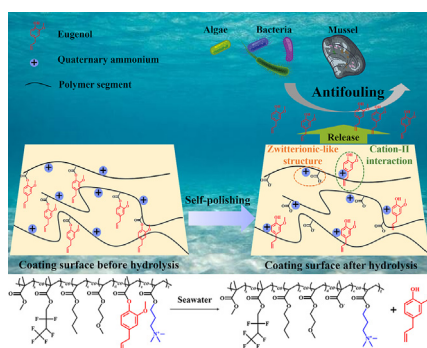
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HIGHLIGHTS

- Eugenol is anchored on the surface via cation- π interaction for the first time.
- The polymer synthesis only requires simple one-step free radical polymerization.
- Significant environmental friendliness compared to traditional coatings.

GRAPHICAL ABSTRACT



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ABSTRACT

Traditional self-polishing antifouling coatings kill surface organisms by releasing toxic substances, which are damaging to the ecosystem. As a natural antimicrobial substance, eugenol is environmentally friendly and has been proven by different research teams to be effective in enhancing the anti-fouling effect of coatings in the real sea. While in these previous research works, the eugenol was released directly into the seawater thus cannot further serve as surface antifouling effect, leading to a limited antifouling effect of the coating. In this work, the quaternary ammonium component was introduced into the butyl ester-based resin - poly (eugenol methacrylate - acryloyloxyethyltrimethyl ammonium chloride - hexafluorobutyl methacrylate - methyl methacrylate - butyl methacrylate - ethylene glycol methyl ether acrylate) (EMQFP) coating for the first time by simple one-step free radical polymerization method. On the one hand, the eugenol produced by hydrolysis is anchored to the quaternary ammonium on the coating surface for a period of time due to the cationic- π interaction, instead of being released into seawater immediately after hydrolysis, thus increasing the utilization rate of eugenol; on the other hand, the negatively charged carboxylate groups generated after hydrolysis in the coating are mutually attracted to quaternary ammonium through electrostatic effect, so the resin chain segment conformation on the coating surface adjusted to produce zwitterionic-like structure, and the hydration of zwitterionic inhibits primary fouling adhesion.

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1. Introduction

Marine biofouling refers to the process by which marine organisms adhere to submerged surfaces, including vessel hulls, pipelines, oil platforms, and offshore sensors, resulting in a significant economic burden on the maritime industry [1–6]. As biofouling raises the roughness of the hull surface and increases the hull mass, it not only increases drag and fuel consumption, but also reduces the maneuverability of the hull and increases the risk of the vessel's navigation [7,8]. Marine antifouling coatings are considered to be one of the most effective means of antifouling [9–14]. The self-polishing antifouling coatings (SPCs) based on hydrolyzable side groups of silicone-based, copper-based or zinc-based acrylate copolymers are the most commonly used because of their efficacy, easy application process and economy [15–17]. However, these traditional SPCs are primarily based on the release of biocides (mainly involving copper and zinc compounds) during hydrolytic polishing to kill organisms on the surface of the coating to achieve antifouling effects [18,19], which causes adverse effects on marine ecology. Many countries and regions around the world have introduced regulations to restrict the use of such antifoulants [20]. New environmentally friendly coatings have become a research trend [13,21,22]. At present, such as superhydrophobic coating materials and nano antifouling coating fillers have been more widely studied [5,23–27]. Although hydrophobic materials exhibit excellent anti-fouling effects, these materials are mostly still in the laboratory research stage, which is difficult to be applied in large quantities in marine coating industry. Therefore, it is important to develop a new strategy to improve the antifouling performance of SPCs as an alternative to traditional heavy metal antifoulants.

Eugenol is researched in marine antifouling owing to its antibacterial properties, biodegradability, and non-toxicity to humans [28–33]. In 2015, M. Pérez et al. [30] found that the incorporation of thymol, eugenol and guaiacol into marine paints showed significant antifouling activity at the concentrations tested. The varnishes containing Cu^+ and eugenol showed similar properties to those containing 16 % copper. The Cu^+ and eugenol-based paint contained 90 % less copper compared to the traditional copper-based paint, achieving similar antifouling effects [31]. In 2020, Xie et al. [34] applied eugenol methacrylate to prepare ternary synergistic antifouling marine antifouling coatings. The coating was proven to be effective in inhibiting the adhesion of proteins, bacteria and boat-shaped diatoms, and the coating can effectively inhibit biofouling in the real sea for at least 8 months. In 2021, a modified eugenol and coumarin (polydimethylsiloxane) PDMS elastomer with antifouling properties was reported by Z. Tong et al. [32]. And a 9-month marine field test was conducted in the

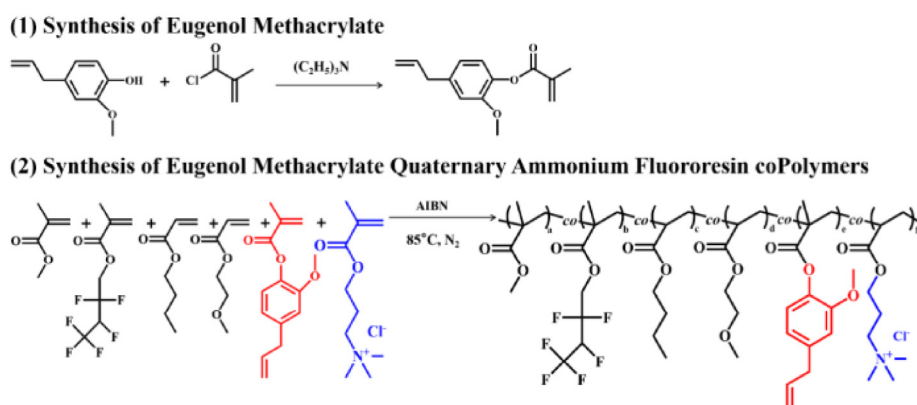
East China Sea near Zhoushan Islands, which proved its good antifouling effect in real sea. In 2022, our team prepared a novel self-polishing antifouling coating polymer that releases eugenol via the hydrolysis of phenolic ester groups [35]. The hydrolytic degradation results show that the degradation of the copolymer is controlled, and the degradation rate increases with increasing contents of eugenol methacrylate (EM). However, in these previous works, the hydrolysis-generated eugenol released into seawater dissipates quickly and thus lacks lasting antifouling effect. As far as we know, currently there is no method to increase the effective utilization of eugenol. And in our previous work, although the antibacterial performance of the coating resin improves with increasing EM content, the improvement extent gradually decreases with increasing EM content. Therefore, the anti-fouling property of the coating cannot be improved by increasing the content of eugenol indefinitely.

Theoretically, the antifouling effect of the coating can be further enhanced if the eugenol is not rapidly released into seawater after hydrolysis but is anchored on the surface for a certain period of time. If this is achievable, the antifouling effect of the coating surface can be improved with a fixed content of eugenol in the coating. For this purpose, quaternary ammonium was introduced into eugenol ester-based methacrylic resins to synthesize the poly (eugenol methacrylate - acryloyloxyethyltrimethyl ammonium chloride - hexafluorobutyl methacrylate - methyl methacrylate - butyl methacrylate - ethylene glycol methyl ether acrylate) (EMQFP) using a simple one-step free radical polymerization method. The synthesis method diagram is shown in Scheme 1. The interaction of the quaternary ammonium cation with the π -structure of the benzene ring of eugenol allowed the eugenol to remain anchored to the coating surface. It is presumed from the GAUSSIAN calculations that the positively charged quaternary ammonium on the coating surface would interact with the negatively charged carboxylate through electrostatic interactions, resulting in a new zwitterionic-like structure of the resin chain segments on the coating surface through conformational adjustment. A significant increase of eugenol was detected on the surface of the coating after hydrolysis. The coating showed obvious inhibition of adhesion towards bacteria, diatoms and mussels.

2. Experimental section

2.1. Materials

Eugenol (AR), methacryloyl chloride/benzoyl (AR), hexafluorobutyl methacrylate (FMA, AR), peroxide (BPO, AR), CaH_2 (AR) and agar powder (BR) and agar powder (BR) were purchased from Aladdin Company. Acryloyloxyethyltrimethyl ammonium chloride



Scheme 1. The synthesis of the EM and EMQFPs via free radical polymerization.

(AAC, 80 wt% in H₂O) was purchased from Tianjin HEOWNS. Azobisisobutyronitrile (AIBN, AR), methyl methacrylate (MMA, AR), butyl methacrylate (BA, AR), ethylene glycol methyl ether acrylate (MEA, AR), triethylamine (AR), KBr (GR) and N,N-Dimethylformamide (DMF, AR) were purchased from Tianjin Guangfu Technology Development Co., Ltd. Ethanol (AR) and D301 ion adsorption resin (AR) were purchased from Tianjin Fuyu Fine Chemical Co. Ltd. Xylene (AR) was from Tianjin Damao Chemical Reagent Factory. D301 was used after normalization. Xylene used for substitution reaction needs to be dried over CaH₂.

2.2. Synthesis of eugenol methacrylate (EM)

Eugenol and triethylamine were pre-dissolved in xylene at a molar ratio of 1:1.2 with continuous stir in ice-water bath to a temperature lower than 5 °C. Methacryloyl chloride with same molar amount as eugenol was slowly added dropwise to a constant pressure dropping funnel. After the completion of the dropwise addition, the reaction was carried out at room temperature for 24 h. The ammonium produced by the reaction was removed by filtration, the unreacted monomer was removed with a cationic adsorption resin, and the yellow EM was obtained after the solvent was evaporated to dryness under low pressure.

The ¹H NMR results indicate the successful synthesis of EM (Fig. 1. (a)). The corresponding relationship between the peak positions of each peak is as follows.

¹H NMR (500 MHz, CDCl₃, ppm): 5.06–5.19 a, b (CH₂CHCH₂-), 5.9–6.1 c (CH₂CHCH₂-), 3.34–3.41 d (CH₂CHCH₂-), 6.80–6.75 e, 6.97–7.00 f, 6.82 g, 3.823 h (CH₃O-), 6.66–6.72 i, 6.42–6.34 j, 2.13–2.02 k CH₃C(CH₂)₂-.

2.3. Synthesis of EMQFPs

EMQFPs were synthesized by free radical polymerization. First, DMF (300 g) was added into a four-necked flask equipped with a condenser, stirrer, thermometer, and dropping funnel. Then the mixture was heated to 85 °C. Subsequently, MMA, BA, FMA, MEA, AAC and AIBN were thoroughly mixed and added to the flask in a uniform slow drop over 3.5 h. After that, DMF (30 g) and BPO (1 g, 0.004 mol) were added to the flask with even dropping within 40 min. Then the reaction was performed with continued stirring for 3 h. Finally, the reaction was terminated by exposing the polymer to air. The dosage ratio and glass transition temperature (T_g) of each monomer are shown in Table 1.

Table 1

Synthetic formula table of the EMQFPs.

Sample	MMA/BA/FMA/MEA/EM /AAC /AIBN/BPO (g)	Mw	PDI
EMQFP-0	154/83/27/6/30/0/4.5/1	2.75×10^4	1.68
EMQFP-1	151/82/27/6/30/3/4.5/1	2.43×10^4	1.41
EMQFP-2.5	149/81/27/6/7.5/30/4.5/1	2.27×10^4	1.21
EMQFP-5	144/78/27/6/30/7.5/4.5/1	2.13×10^4	1.44
EMQFP-10	134/72/27/6/30/15/4.5/1	1.94×10^4	1.17

3. Results and discussion

3.1. Synthesis of EM and EMQFPs

A new peak at 1268 cm⁻¹ for EM corresponds to stretching vibration of C—O—C, demonstrating the formation of ester groups (Fig. 1 (b)). The FT-IR spectra of the fluoro/silyl acrylate copolymer are shown in Fig. 1 (c), where the peak at 3000–2850 cm⁻¹ corresponds to the stretching vibration of C—H in methyl and methylene, the peak at 1734 cm⁻¹ corresponds to the stretching vibration of carbonyl in methyl methacrylate, and the peak at 1265 cm⁻¹ corresponds to the stretching vibration of C—F. The peak near 1610 cm⁻¹ corresponds to the stretching vibration of C=C. 1250–1030 cm⁻¹ corresponds to the stretching vibration of C—N. 2800 cm⁻¹ corresponds to the stretching vibration of CH₃ linked to N. Due to the presence of tertiary amine structures in the solvent DMF, which cannot be completely evaporated, and the presence of incomplete triethylamine in the resin, the tertiary amine structures present in EMQFP-0 also show peaks at 1250–1030 cm⁻¹ and 2800 cm⁻¹. ¹H NMR and FT-IR results indicated the successful synthesis of EM and EMQFP.

3.2. Hydrolysis properties

Self-polishing coatings must have stable polishing rates in practical applications. The polishing of self-polishing antifouling coatings is achieved by hydrolysis of the EMQFP resin. After the hydrolysis of the phenolic ester group, more hydrophilic -COO⁻ is exposed on the coating surface, which makes the resin on the coating surface brittle and peeling off. The EMQFP coatings were immersed in artificial seawater for 30 days for weight loss measurements (Fig. 2 (a)). It was observed that the coatings exhibited stable hydrolysis rates in artificial seawater, and the rate of hydrolysis weight loss accelerated with the increase of quaternary

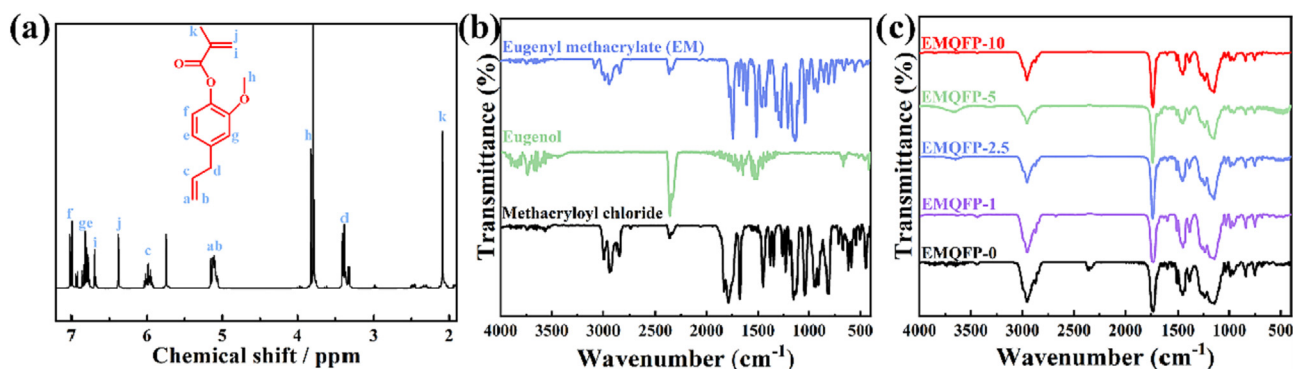


Fig. 1. (a) ¹H NMR spectra of EM (b) FT-IR spectra for EM, eugenol and methacryloyl chloride. (c) FT-IR spectra for EMQFPs.

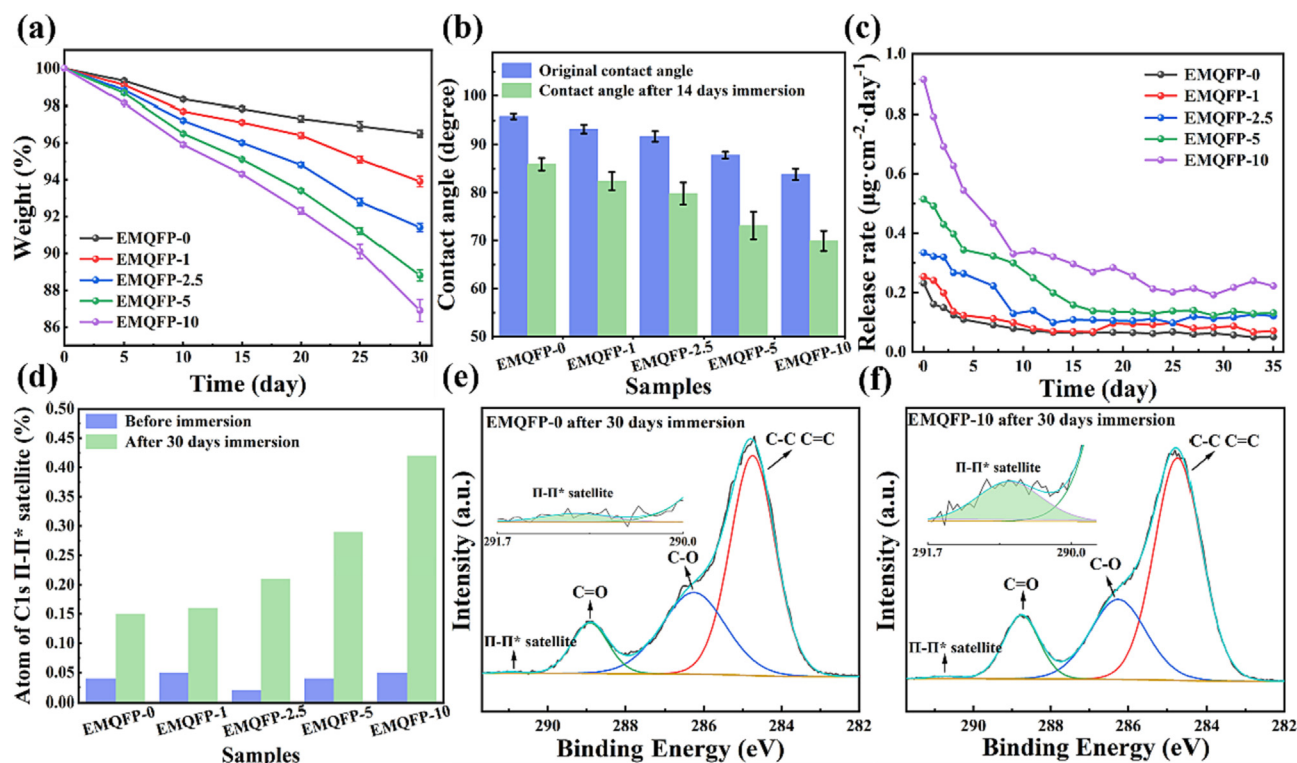


Fig. 2. (a) Weight loss ratios of EMQFP coatings. (b) The contact angle of EMQFP coatings before and after immersion in artificial seawater for 14 days. (c) Release rate of eugenol from EMQFP coatings in artificial sea water at 25 °C. (d) Atom of C1s π - π^* satellite content in EMQFP coatings. High resolution C1s spectra of EMQFP-0 (e) and EMQFP-10 (f) after 30 days of immersion in artificial seawater.

ammonium content. The remaining mass of EMQFP-10, which lost the most weight after 30 days of immersion, was $86.9 \pm 0.6\%$, while the remaining mass of EMQFP-0, which lost the least weight, was $96.5 \pm 0.2\%$. The mass loss rate of EMQFP-10 was approximately 3.7 times higher than that of EMQFP-0. It is because the quaternary ammonium component is a hydrophilic functional group, which accelerates hydrolysis. This point is supported by Fig. 2. (b), as the contact angle of the coating before hydrolysis decreased with increasing quaternary ammonium content. After 14 days of immersion, the hydrophilicity of the coating surface increases. On the one hand, the hydrolysis of eugenol esters on the coating surface produces hydrophilic carboxylate functional groups [35]; on the other hand, the hydrophilic functional group (quaternary ammonium) would migrate to the coating surface due to the conformational adjustment of chain segments on the coating surface [36,37].

The release rate of eugenol was measured by the UV spectrophotometer (Fig. 2 (c)). It is observed that although the content of eugenol fraction in the coating resin is the same (Table 1), the eugenol release rate increases with the increase of quaternary ammonium fraction. The eugenol release rate was the fastest at the beginning of immersion. The release rate decreases with increasing immersion time in artificial seawater. After 35 days, the release rates of eugenol of EMQFP coatings were all stable. The fastest release rate was $0.22 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$ for EMQFP-10, while the slowest release rate was $0.05 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$ for EMQFP-0. This is attributed to the reason that the faster the rate of hydrolysis weight loss, the faster the rate of eugenol release. To study the changes in the eugenol content of the coating surface, EMQFP samples before and after immersion in artificial seawater for 30 days were characterized by XPS measurements. The relative content of benzene ring carbon and aliphatic chain carbon on the surface of the dried coating was evaluated to reflect the content of eugenol on the surface of the coating. The π - π^* satellite repre-

sents the benzene ring carbon in the resin, that is, the content of eugenol. It can be seen that the content of eugenol on the surface of the coatings after 30 days of immersion is higher than that on the surface of the coatings before hydrolysis (Fig. 2 (d) and Table S1.). The difference in the surface eugenol content of EMQFP coatings before immersion was not significant, but after immersion, the surface eugenol content apparently increased with the increase in quaternary ammonium content. This confirms that the presence of quaternary ammonium can indeed increase the eugenol content on the surface of the hydrolyzed EMQFP coatings.

The possible reason is that there is a cation- π interaction between the cation of the quaternary ammonium and the π -bonded structure of the eugenol [38,39], and the cation of the surface layer is able to anchor the partially hydrolyzed detached eugenol to the surface by this weak interaction force instead of diffusing it directly into the seawater. On the other hand, in the coating, the eugenol component on the resin chain segment has cation- π interaction with the quaternary ammonium component as well, and as the quaternary ammonium component migrates more to the surface of the coating, the eugenol ester structure may also migrate to the surface with it. To verify this, GAUSSIAN was used to calculate the interaction potential energy of the positively charged center N^+ of the quaternary ammonium with the eugenol structure and the carboxylate produced after hydrolysis. The calculated interaction potential energy of N^+ with eugenol is -22.98 kcal/mol , while the interaction potential energy with carboxylate is -100.55 kcal/mol . The calculations proved that the eugenol produced by hydrolysis does interact with the quaternary ammonium, but the nitrogen atom center N^+ of the quaternary ammonium is more likely to interact with the carboxylate exposed by hydrolysis through electrostatic interactions. This suggests that the carboxylate produced after hydrolysis may interact with the quaternary ammonium salt to produce a zwitterionic-like structure.

3.3. Surface morphology of EMQFP coatings

The surface stability of the coating must be maintained in actual use, so it is important to characterize the surface micromorphology of the coating before and after immersion in artificial seawater. The surface morphology of the coating before and after 14 days of immersion was observed by SEM, and the nanoscale microstructure and roughness of the surface after 14 days of immersion were measured by AFM. The surface of the coating was flat without wrinkles or cracks before immersion (Fig. 3 (a-e)), demonstrating the good film-forming properties of EMQFP coating. After 14 days of immersion in artificial seawater, micro-folds appeared on the surface of the coating, but the coating as a whole remained intact without cracks (Fig. 3 (f-n)), indicating the stability of the coating. Microfolds on the coating surface increased with increase of quaternary ammonium fraction. The roughness of the coating surface at the nanoscale observed by AFM increases accordingly. This may be due to two reasons. On the one hand the quaternary ammonium component is a hydrophilic functional group which led to a rapid local hydrolysis of the coating. On the other hand, the presence of cationic- π interaction between the quaternary ammonium and eugenol ester functional groups led to conforma-

tional changes of the chain segments on the surface. Overall migration of resin chain segments resulted in the appearance of microfolds.

3.4. Antifouling performance test on EMQFP coatings

Bacteria and diatoms are common primary fouling microorganisms. Primary colonization by these microorganisms produces a biofilm matrix onto which other marine fouling organisms adhere [40,41]. To investigate the antimicrobial properties of the eugenol release steady state, the EMQFP coating was immersed in artificial seawater for 35 days before being tested.

The antibacterial performance of the EMQFP coatings was investigated using *E. coli* (Gram-negative bacteria), *S. aureus* (Gram-positive bacteria) and *Pseudoalteromonas marina* as model bacteria. The antibacterial efficiency of EMQFP coatings was calculated using EMQFP-0 as the control group, and the results are shown in Fig. 4. (b). The antibacterial efficiency of EMQFP coating against all three bacteria increased with the increase of quaternary ammonium fraction. Among them, the antibacterial efficiency of EMQFP-10 against *E. coli*, *S. aureus*, and *Pseudoalteromonas marina* was $87.7 \pm 3.5\%$, $89.9 \pm 3.7\%$, and $91.1 \pm 1.9\%$, respectively. This

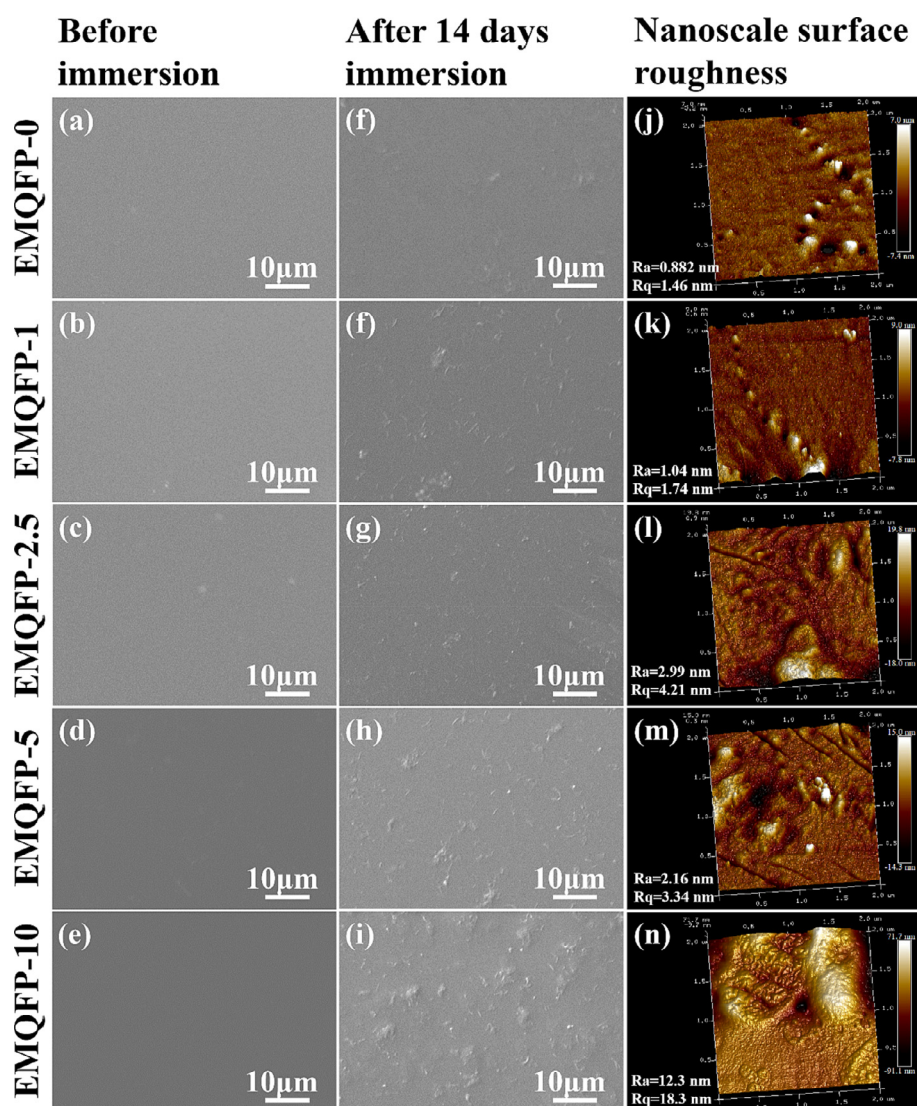


Fig. 3. Representative SEM images of EMQFP coatings (a-e) before immersion and (f-i) after 14 days immersion in artificial seawater; and (j-n) representative AFM images of EMQFP coatings.

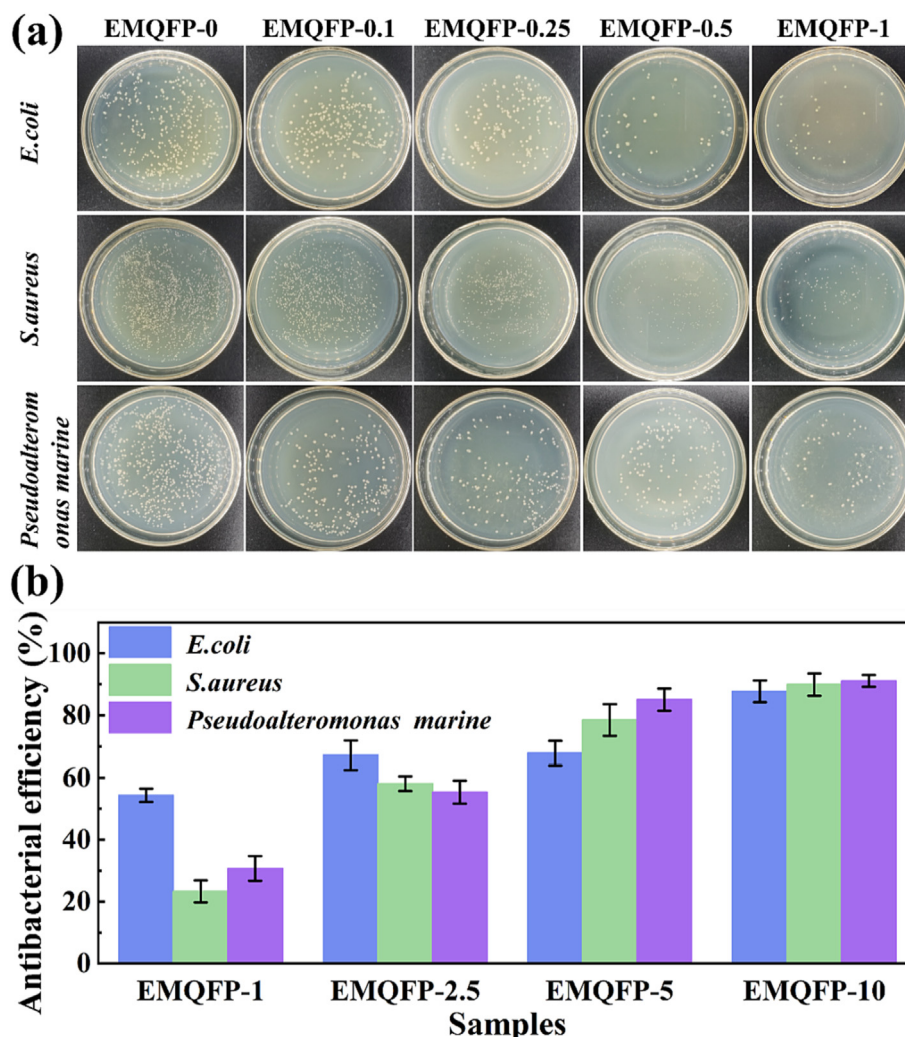


Fig. 4. (a) The digital photographs of the colonies incubated on agar plates with different EMQFP coatings toward *E. coli*, *S. aureus* and *Pseudoalteromonas marina*, (b) Antibacterial efficiency of EMQFP coatings.

proves that the presence of quaternary ammonium fraction can improve the wide-spectrum antibacterial ability of the coating. The excellent antimicrobial properties may be due to three aspects. Firstly, due to the hydrophilic property of the quaternary ammonium fraction, the quaternary ammonium would migrate to the coating surface by conformational changes, resulting in an increase in the quaternary ammonium fraction of the coating surface and thus an increase in antibacterial ability. Secondly, due to the cation- π interaction, some of the eugenol that is hydrolyzed out of the polymer was anchored on the coating surface. Eugenol can denature proteins and react with cell membrane phospholipids to change their permeability, thus inhibiting a great number of Gram-negative and Gram-positive bacteria [42]. Finally, hydrolysis of the phenolic ester group leaves negatively charged carboxylate on the polymer chain segment, which interacts electrostatically with the positively charged quaternary ammonium. The polymer forms a surface zwitterionic carboxybetaine-like structure by adjusting the conformation. The carboxybetaine-like structure possesses hydration, which inhibits bacterial adhesion by forming a water shield on the coating surface to inhibit contact between bacteria and the material surface [43–46].

P. tricornutum (plankto diatom) and *Halamphora* sp. (benthic diatom) were selected as representatives to further evaluate the broad-spectrum anti-diatom adhesion properties of EMQFPs coat-

ing. The anti-diatoms ratio of EMQFP coating was calculated using EMQFP-0 as the control group, and the results are shown in Fig. 5. (b, c). The results showed that the coating had an inhibitory effect on adhesion for both diatoms. The effect of adhesion inhibition enhanced with the increase of quaternary ammonium fraction. It is presumed that the main reason for inhibiting diatom adhesion is that the quaternary ammonium and the carboxylate produced after immersion formed a zwitterionic-like structure through electrostatic attraction, and the surface hydration layer inhibited the direct contact between diatoms and the coating surface. By calculating the number of diatom on the surface, the inhibition ratios of EMQFP-1, EMQFP-2.5, EMQFP-5 and EMQFP-10 against *P. tricornutum* (plankto diatom) after 24 h immersion are 9.9 ± 0.9 %, 22.7 ± 0.9 %, 14.3 ± 2.8 %, and 33.7 ± 4.0 %, respectively. Comparably, the inhibitory ratios after 48 h immersion are much higher, at 23.7 ± 3.1 %, 32.2 ± 3.5 %, 36.3 ± 4.1 %, and 52.0 ± 4.7 %. It can be concluded that the longer the immersion time, the obvious advantage of the samples with quaternary ammonium fraction against *P. tricornutum* (plankto diatom) adhesion. The reason may be that the quaternary ammonium on the surface of EMQFP coatings and carboxylate electrostatically attracted to adjust the conformation to form zwitterionic-like structure, while no similar structure was formed on the surface of EMQFP-0 coating. The higher *P. tricornutum* concentration after 48 h due to the diatoms being in the

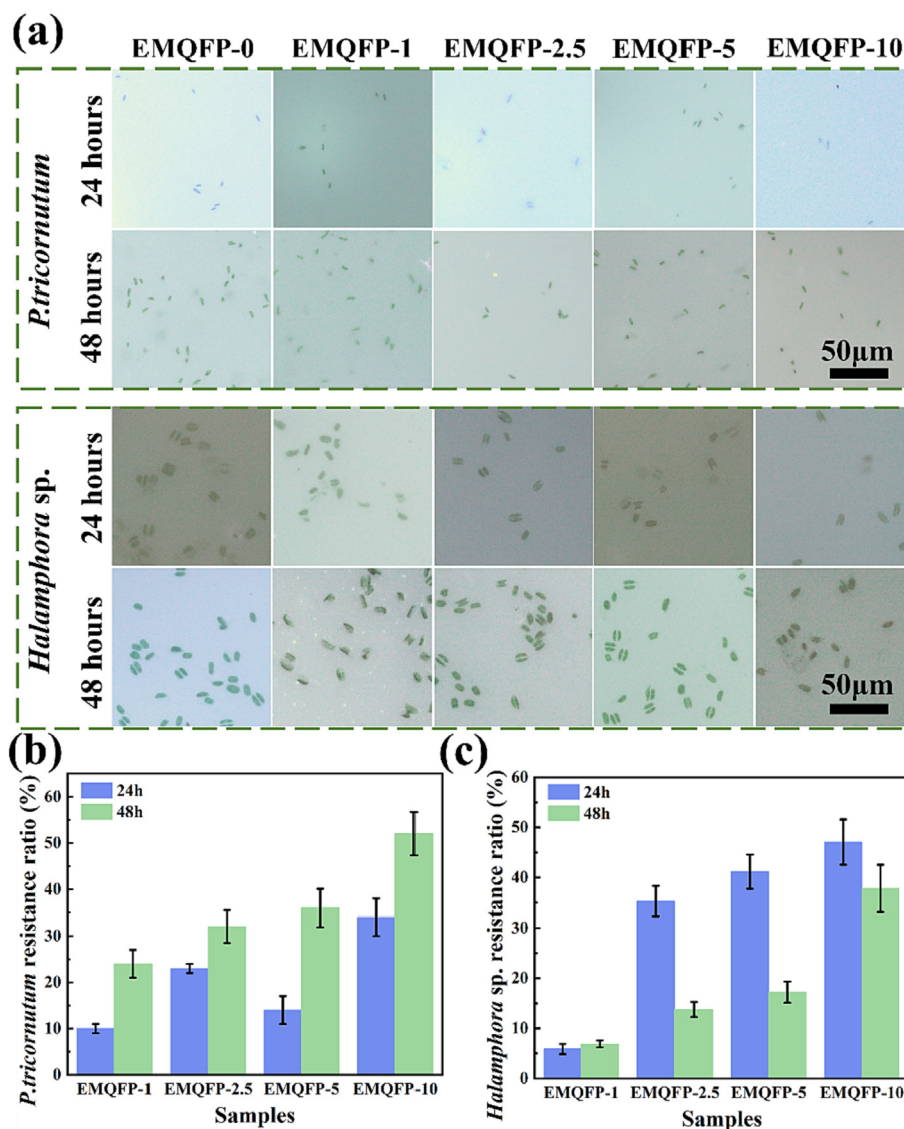


Fig. 5. (a) Representative fluorescence microscopy images of different EMQFP coatings immersed in *P. tricornutum* and *Halamphora* sp. solutions after 24 h and 48 h (b) *P. tricornutum* resistance ratio and (c) *Halamphora* sp. resistance ratio of EMQFP coatings after 24 h and 48 h immersion.

growth phase resulted in more *P. tricornutum* adhesion on the EMQFP-0 surface, while the amount of newly adhered *P. tricornutum* on the zwitterionic-like surface due to hydration was much lower than on the EMQFP-0 surface (Fig. 5 (a)). The inhibition ratios of EMQFP-1, EMQFP-2.5, EMQFP-5 and EMQFP-10 against *Halamphora* sp. (benthic diatom) after 24 h immersion are 5.9 ± 1.0 %, 35.3 ± 2.9 %, 41.2 ± 3.4 %, and 47.0 ± 4.5 %, respectively. Comparably, the inhibitory ratios after 48 h immersion are much higher, at 6.9 ± 0.7 %, 13.8 ± 1.5 %, 17.2 ± 2.1 %, and 37.7 ± 4.8 %. While EMQFP coatings have very different adhesion inhibition effects on *Halamphora* sp. Inhibition of adhesion by EMQFP-1 to *Halamphora* sp. was slightly better after 48 h of immersion than after 24 h. The remaining samples were significantly less resistant to diatom adhesion after 48 h of immersion than after 24 h of immersion.

The possible reason is that the coating surface has microstructures (Fig. 3), which, according to the adhesion point theory [47,48], increased the contact area between the benthic diatom and the coating surface, and therefore the longer time was favorable for the adhesion of *Halamphora* sp. to the coating surface. While in the real marine environment, the ship surface is mainly

in contact with plankto diatom, so the EMQFPs coating has the potential to reduce the adhesion of diatom in the real marine environment. Therefore, the addition of the quaternary ammonium component to the EMQFP coating may have an inhibitory effect on macrofouling bio-adhesion.

Mussels are typically large fouling organisms which lead to larger harm than diatoms [49]. *Mytilus edulis* as a model species was used to study the anti-mussel performance of the EMQFPs because it is one of the most common marine bio-fouling [50–52]. An assay of the test panel, which was composed of different samples, is shown in Fig. S1. At the beginning of the test, the mussels were evenly placed on the mosaic test board, in which different samples were spliced together. Three groups of repeat experiments were conducted to assess the effects of different EMQFP samples on selective attachment of Mussels. Digital photos of the board initially and after 24 h are shown in Fig. S1. From the group 1, it can be seen that one mussel adhered to the surface of EMQFP-2.5, one adhered to the junction of EMQFP-0 and EMQFP-1, two adhered to the glass, one adhered to the surface of EMQFP-0, one adhered to the gap between EMQFP-10 and EMQFP-2.5, and two moved outside the test area; From the second group it can be seen

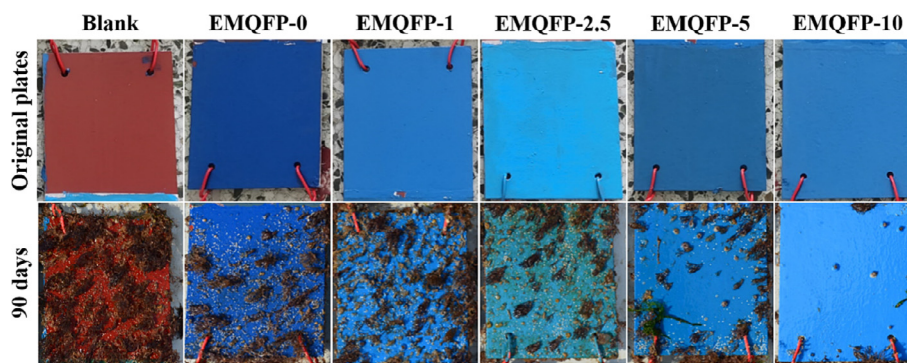


Fig. 6. Photographs of the EMQFP coatings and blank immersed in the Yellow Sea of China for 90 days. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

that three mussels were adhered to each other, the bottom one adhered to the empty glass, one mussel adhered to the gap between EMQFP-0 and EMQFP-5, one mussel adhered to the empty glass, and three other mussels moved outside the test area; From the third group, it can be seen that two mussels adhered to the glass surface, one adhered to the surface of EMQFP-5, one did not secrete foot filaments to stay in place, two mussels moved to the edge of the test area foot filaments adhered to the plate without adhering to the sample surface, one adhered to the gap between the two plates, and one adhered to the mussel shell and EMQFP-2.5 at this gap. Overall, it appears that mussel selection experiments proved the mussel's tendency to avoid EMQFP coatings with high quaternary ammonium content.

To explore the antifouling performance of EMQFP coatings in an actual marine environment, marine antifouling experiments were carried out in the Yellow Sea of China (photographs of the coated and blank plates are shown in Fig. 6). It can be seen that the blank sample was covered with fouling organisms after 90 days of immersion. The fouling organisms on the surface of EMQFP-0 were significantly less than those on the blank samples, proving the butyl ester resin coating can indeed perform marine antifouling effect. With the increase of the quaternary ammonium fraction, the fouling organisms on the surface of the coating decreased significantly, and only a few barnacles were found on the surface of the coating EMQFP-10. The results indicated good antifouling performance for the EMQFP coatings. The experiments on marine pegboard proved that the increase of quaternary ammonium fraction in butyl ester resin did improve the antifouling effect of the coating in an actual marine environment.

4. Conclusions

This work is based on the fact that eugenol has excellent marine antifouling properties [30–32,34] and the phenol ester-based hydrolytic self-polishing antifouling coating technology [35] reported in our previous work. This work achieved the concentration of eugenol on the coating surface for the first time, significantly improving the efficiency of eugenol utilization and improving the antifouling effect. The quaternary ammonium component may anchor the eugenol released by the hydrolysis of the phenolic ester group to the surface via cation- π interaction. The carboxylate produced by hydrolysis may be mutually attracted with the quaternary ammonium fraction through electrostatic interaction, and the polymer chain segments on the surface of the coating form an zwitterionic-like structure through conformational adjustment, forming a hydrated layer on the surface of the coating. The presence of the quaternary ammonium component improves the coating's resistance to diatom adhesion and bacterial adhesion, and the mussel has a tendency to avoid the high quater-

nary ammonium content of the butyl ester-based resin. The coating has been tested on real sea pegboards and confirmed to have a real marine antifouling effect. Since the hydrolysis of phenolic ester groups to produce phenolic hydroxyl groups is universal for all natural phenolics, and since phenolics naturally contain a benzene ring π -bond structure, this means that a wider variety of phenolic compounds could potentially be applied to self-polishing antifouling coatings and improve antifouling efficiency through this cation- π interaction. This work is expected to provide new ideas for the industrial production of environmentally friendly antifouling coatings.

CRediT authorship contribution statement

Jianang Sha: Conceptualization, Experiment, Writing- Original draft preparation. **Xin Liu:** Data curation. **Rongrong Chen:** Writing – review & editing, Investigation. **Jing Yu:** Writing – review & editing. **Qi Liu:** Investigation, Supervision. **Jingyuan Liu:** Detection, Validation. **Jiahui Zhu:** Detection, Software. **Peili Liu:** Data curation, Validation. **Rumin Li:** Software, Investigation. **Jun Wang:** Conceptualization, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2023.01.058>.

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